

Plan — Reward-Driven TD3 Dynamic Capture (Stage 2)

An action-preserving residual over the frozen phase-shaped scaffold; the claim is robustness, not re-derivation

HyMeKo coin-delivery campaign (recovery/coin-r9-causal-residual-delivery)

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Abstract

Stage 1B proved a structured phase-shaped moving-precursor capture delivers strict K6 from HOME_V1 (3/3 seeds, frozen downstream). This RL is genuinely justified: not re-learning known kinematics, but turning a *proven, reachable* moving contact-handoff into a state-dependent, *robust* capture policy. The critical scientific rule: a TD3 that merely reproduces the canonical-start CEM profile proves nothing (the scaffold already solves it). The claim is TD3_IMPROVES_OVER_FROZEN_SCAFFOLD_PASS — the learned policy must beat both the frozen scaffold *and* a BC-only checkpoint on a *pre-registered* perturbation panel (small admissible READY / coin-state / actuator-state variations), with no safety regression. We freeze one scaffold π_0 by an *outcome-independent* rule (structural-parameter medoid), wrap it action-preservingly ($u = \text{clip}(u_{\pi_0}(\phi) + \alpha \tanh \delta_\theta(s), -1, 1)$; zero residual $\equiv \pi_0$, bit-exact, tested before training), and train a genuine reward-driven TD3 residual over only READY→capture→REACH_TO_APPROACH — everything else frozen. *Learn the recovery, not the trajectory.*

1 Scope and goal

Goal. A learned TD3 residual capture policy, deployed with no CEM/teacher in the loop, that (i) delivers teacher-free canonical HOME_V1→strict K6 and (ii) *improves robustness* over the frozen scaffold and BC-only on a pre-registered development perturbation panel. **Deployed chain:**

*HOME_V1 → frozen analytic transit → learned TD3 dynamic capture → frozen APPROACH →
HANDOFF_RESET → frozen R2 → strict K6.*

Non-goals (explicit stops). No full multiseed panel; no SAC/PPO comparison; no Hamiltonian/energy/entropy experiments. No modification of the frozen downstream, the analytic transit, or physics/safety/K6. This is the bounded first campaign only.

2 The frozen contract

- Frozen and imported as-is: HOME_STATE_V1, the analytic HOME→READY transit (eb09a16a), the Stage-1B scaffold machinery (moving_precapture, d6974b95), APPROACH / HANDOFF_RESET / R2 / coast, and physics / safety / strict-K6. `git diff` empty on all.
- TD3 controls *only* READY→DYNAMIC_CAPTURE→REACH_TO_APPROACH, via the real 4-D slew-normalised *residual* action, through the governed step_ablation stack. No teacher / future info; no state edit; no online CEM once training begins.
- s4/s7 validation-only; f1--f4 SEALED. Development seeds only. No tag moved.

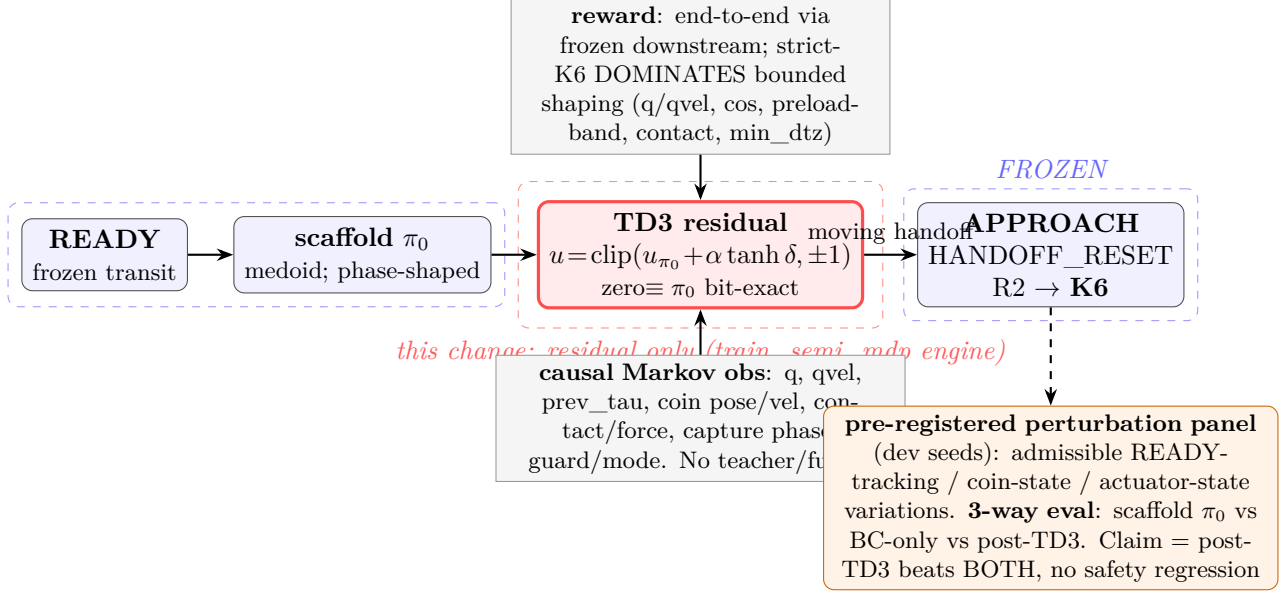


Figure 1: The TD3 residual (red) wraps the frozen scaffold π_0 (blue) action-preservingly; training scores the full end-to-end rollout through the frozen downstream. The learning claim is measured on a pre-registered perturbation panel against scaffold and BC-only baselines.

3 Affected files

CORE.YAML items touched: none. Reuse (read-only): option_rl.agents (train_semi_mdp, make_actor, OptionEnv/OptionTransition, SemiMDPCConfig — the dim-generic TD3/SAC engine); kinetic_authority_u (zero_init_detactor, deterministic_residual, champion_key_unlock, expl_noise_for); moving_precapture (scaffold). **New:**

- coin_delivery/theta_option/capture_rl.py — freeze_scaffold (medoid selection), ResidualCaptureController (action-preserving wrapper), capture_observation (causal Markov state), capture_reward (K6-dominated bounded shaping), CaptureOptionEnv (reset/step over the frozen scaffold + downstream), PerturbationPanel (pre-registered), the 3-way evaluate + BC init.
- experiments/coin_kinetic_capture_td3.py — identity test $\rightarrow$ BC $\rightarrow$ 3-seed TD3 $\rightarrow$ 3-way panel eval $\rightarrow$ gates $\rightarrow$ provenance/traces.
- tests/test_capture_rl.py; reports/2026-07-28-td3-capture-reward-driven.md.

4 Interface changes

- freeze_scaffold(results) $\rightarrow$ CaptureParams — the structural-parameter *medoid* of the K6 solutions (outcome-independent; *not* best min_dtz). **Post:** deterministic; documented rule.
- ResidualCaptureController(ready, ref, pi0, residual_fn, alpha) — rolls the scaffold with $u = \text{clip}(u_{\pi_0}(\phi) + \alpha \tanh \delta(s), -1, 1)$. **Post:** residual_fn $\equiv 0$ reproduces π_0 *bit-exact* (asserted by IDENTITY test).
- CaptureOptionEnv — reset() $\rightarrow$ s; step(a) $\rightarrow$ (s2, r, done, info); on the terminal capture step, rolls the frozen downstream and adds the strict-K6-dominated terminal reward. info carries k6, safe, min_dtz, boundary counts.
- evaluate(actor|pi0, panel) $\rightarrow$ metrics — K6 rate + Wilson CI, handoff margins, safety, on the pre-registered panel; used identically for scaffold / BC-only / post-TD3.

5 Test strategy

Unit: medoid selection determinism + outcome-independence; `capture_observation` schema + no future/teacher leak; reward monotonicity (K6 dominates shaping); panel determinism. **Integration/identity (load-bearing):** `RESIDUAL_ZERO_IS_SCAFFOLD_IDENTITY` — zero residual reproduces π_0 's rollout *bit-exact*; the wrapped scaffold at $\alpha=0$ delivers the same K6/min_dtz as `moving_precapture`. **Smoke (production scale, 1 seed):** a short TD3 run improves panel K6 over BC-only/scaffold (else **HALT** and localise — reward, obs, or exploration). **Boundary:** exactly-one `READY`→`CAPTURE` / `REACH_TO_APPROACH` / `HANDOFF_RESET`; deterministic replay. **Coverage:** every new function exercised.

6 Performance budget

One capture episode + downstream ≈ 0.1 s; `total_options` ≤ 600 /seed, eval every 25; 3 seeds ≈ 10 –20 min, *checkpointed per seed*. **Peak RSS** < 2 GB (torch actor/critic; hard cap 16 GB). Unit suite < 180s. The run reports peak RSS + wall on exit; wall-estimate reconciled before launch (per the $> 2\times$ halt rule).

7 Risk anticipation

Performance-contract preservation. The residual is bounded ($\alpha \tanh$, then the clip to $[-1, 1]$ and the $\times \text{slew} \rightarrow \text{governor}$); it cannot exceed the slew/torque/joint-speed caps. Zero residual is bit-exact π_0 , so the worst case is never worse than the proven scaffold. **The scientific trap.** Canonical-start K6 is *necessary but not sufficient* — π_0 already passes it. The reward-driven claim rides entirely on the pre-registered perturbation panel and the 3-way (scaffold / BC-only / post-TD3) comparison; the panel + selection rule are fixed *before* any TD3 run and never tuned to results. **Reward hygiene.** Terminal strict-K6 dominates the bounded shaping so the policy cannot farm a “good handoff score / bad downstream” side-path (the failure mode seen in Stage 1B coordinate-matching). **Worst-case input:** admissible `READY` tracking error + coin-state + actuator-state perturbations (the panel) — the regime where the fixed scaffold is expected to degrade and a state-dependent policy should recover.

8 Rollback path

New files only; rollback = delete. Zero-residual identity guarantees a revert to π_0 (d6974b95) with no behaviour change. Each checkpoint (`update0/BC/best`) is saved; a bad post-TD3 reverts to BC-only or scaffold.

9 Gate ladder + stop logic

1. `RESIDUAL_ZERO_IS_SCAFFOLD_IDENTITY_PASS` (before training).
2. `HOME_V1_START_STRICT_K6_PASS`: post-TD3, teacher-free canonical `HOME`→K6 (necessary).
3. `TD3_CAPTURE_REWARD_DRIVEN_PASS`: reward-driven improvement over the BC-only checkpoint.
4. `TD3_IMPROVES_OVER_FROZEN_SCAFFOLD_PASS`: post-TD3 beats *both* π_0 and BC-only on the pre-registered panel (higher K6 rate / better handoff margin), no safety regression — *the learning claim*.

Stop logic. Freeze the *first* checkpoint meeting all of: teacher-free canonical K6; improvement over BC-only; improvement over scaffold on the panel; no safety/stall/clamp/reversal; exactly-one each boundary transition; deterministic replay. Then commit and **HALT** — no full multiseed, no SAC/PPO, no energy experiments. If the panel improvement fails, **HALT** and localise (reward shaping, observation, exploration), do *not* widen scope.

10 Empty-plan-dir hygiene

If abandoned before the module lands, delete `docs/plans/2026-07-28-td3-capture-reward-driven/` this session.